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Ultrametric Quantum Kernels: Exact p -adic Feature Maps for Hierarchical Data

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Abstract

Written last, once every number is final.

Keywords

quantum machine learning, quantum kernels, ultrametric spaces, p -adic numbers, hierarchical representation learning

1. Introduction

Draft pending.

2. Preliminaries

Draft pending.

3. Ultrametric Quantum Kernels

Draft pending.

4. A No-Go Theorem for Product Encodings

Draft pending.

5. A Dimension Lower Bound for Arbitrary Encodings

Draft pending.

6. Exact Realisation by Path States

Draft pending.

7. Beyond Exactness: Re-uploading and Noise

Draft pending.

8. Experiments

Draft pending.

9. Related Work

Draft pending. The citations below are placed so that every verified entry is exercised by the build; prose replaces them in Task 17.

10. Limitations and Outlook

Draft pending.

11. Conclusion

Draft pending.

Declaration on Generative AI

Complete per the CEUR-WS generative-AI disclosure policy in force at submission.

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